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A critical analysis of marine carbon sequestration opportunities in South Korea

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Abstract

South Korea has made significant commitments to pursuing marine carbon sequestration [including 'blue carbon'] initiatives as part of its broader environmental and climate strategies. Specifically, the South Korean government has set a target to sequester 1,362,000 tonnes of CO_2 in the ocean by 2050 as part of its national strategy. Here, leveraging available data, we outline potential measures to achieve this goal, and provide critical insights into the scale and feasibility of marine carbon sequestration initiatives to inform policymakers and industry stakeholders. We investigated a wide range of potential approaches, ranging from traditional blue carbon approaches involving conservation and restoration of seagrass meadows and tidal marshes; to emerging strategies involving seaweed farming and mudflat restoration; to geoengineering interventions involving ocean alkalinity enhancement. Overall, we find that the South Korean Government target is achievable, largely through [in order of low to high abatement scaleability]: mudflat and saltmarsh conservation/restoration, seaweed conservation/restoration, seagrass conservation/restoration, seaweed farming and ocean alkalinity enhancement. However, we stress that our estimates are rudimentary and carry numerous assumptions/risks, and, moreover, carbon offset standards are still under consideration and development for some of these abatement approaches. In terms of 'readiness to implement', South Korea is strongest in seaweed carbon sequestration research and application, with a track record of successful restoration of tens of thousands of hectares of seaweed habitats over several decades. A coordinated national strategy will be needed to realise and establish South Korea's marine carbon sequestration potential, supported by policy and finance. Fortunately, the marine carbon strategies proposed align with the

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country's broader initiatives to enhance biodiversity, protect coastlines, and mitigate the impacts of climate change.

1. Background

Carbon fixation in the ocean ['marine carbon fixation'] accounts for at least 50% of global carbon fixation [1] and therefore has attracted significant attention globally as a potential means to lower atmospheric carbon emissions. Many nations around the world are actively pursuing marine carbon sequestration initiatives as part of their climate mitigation and environmental conservation efforts, including South Korea. The South Korean government has a target to sequester 1,362,000 tonnes of CO_2 via marine carbon sequestration [particularly through 'blue carbon'] by 2050 as part of their national blue carbon strategy, around 0.20% of South Korea's CO_2 emissions in 2013 [2]. However, it is currently unclear how this target will be met as a national assessment of the opportunities for marine carbon sequestration in South Korea has not yet been released. Hence, a systematic investigation into the range of potential marine carbon sequestration opportunities, including their potential scale and feasibility, will be an important step forward toward setting targets.

South Korea is a Peninsula, 91% of which is bordered by the sea and has an Economic Exclusive Zone that is approximately 5 times the total land area, putting South Korea in a good position to capitalize on marine carbon sequestration opportunities. South Korea has seagrasses, saltmarshes, mudflats, extensive seaweed forests and seaweed aquaculture [3–6]. While there are no mangroves in South Korea, there have been suggestions to introduce them for carbon sequestration and coastal protection, as climate conditions may become suitable to support mangrove growth.

Most of the global focus on actionable marine carbon sequestration has been around the coastal ecosystems seagrasses, mangroves and salt marshes, known as 'Blue Carbon Ecosystems' [7]. It has been estimated that there are 380,000 ha covered by South Korea coastal blue carbon ecosystems [seagrasses and saltmarshes], which could potentially sequester approximately 1,010,000 t of CO_2 [8]. However, these estimates do not represent the additionality required to contribute to climate mitigation through avoiding emissions or increasing sequestration through conservation and restoration of blue carbon ecosystems. Moreover, there are several other emerging strategies for marine carbon sequestration, such as restoring seaweed forests [9] as well as geoengineering approaches such as ocean alkalinity enhancement. Hence, there is a need to assess opportunities for marine carbon sequestration in support of the South Korea's intentions to increase marine carbon sequestration. South Korea is specifically focused on three components of Blue Carbon;

1. *Enhancing the carbon sequestration capacity and climate disaster response capability of the oceans*: The goal is to increase carbon sequestration through marine vegetation, aiming to significantly expand the current area of salt marshes

and seaweeds, and to create marine forests using seagrass and seaweeds, targeting an 85% increase in the current area by 2030. Additionally, restoring abandoned salt farms or aquaculture sites to salt marshes to restore carbon sequestration and designating more than half of all salt marshes as protected areas are efforts to maintain their value as carbon sinks.

2. *Expanding participation in blue carbon creation through private, local, and international cooperation*: This involves linking corporate ESG [Environmental, Social, and Governance] management, forming blue carbon partnerships, and establishing a basis for participation by fishermen and local governments to expand participation in blue carbon creation. It also includes securing international carbon accreditation through international carbon credit standards.

3. *Establishing a new blue carbon certification and a long-term promotion foundation*: This focuses on intensive research on the carbon sequestration of new blue carbon candidates and aims to include seaweed in the Intergovernmental Panel on Climate Change [IPCC] guidelines for national sequestration reporting through consensus-building within the international community. Additionally, the strategy targets the creation of research infrastructure by sea area, surveys of area changes in coastal wetlands by source and sink, and the improvement and sophistication of greenhouse gas statistical calculation methods.

In addition to government support, blue carbon is met with growing interest by industry [e.g., car manufacturers] and the public in South Korea, as a recent survey of public perceptions of blue carbon suggested that the social values of blue carbon projects outweigh the costs of restoration [10]

Here we provide an overview of the key opportunities, risks, and potential scale for additional marine carbon sequestration to help inform policy makers and industry on the potential for marine carbon sequestration potential in South Korea. We do so based on the best available data to inform first-order estimates on the scale of opportunity for marine carbon sequestration in South Korea. We also discuss financing mechanisms to create additional blue carbon resources in South Korea and the next key steps to scale up marine carbon sequestration. We assess here potential marine carbon sequestration opportunities in South Korea from the following strategies: saltmarsh conservation and restoration, seagrasses conservation and restoration, mudflats conservation and restoration, seaweed aquaculture and afforestation, and geoengineering via ocean alkalinity enhancement [depicted in Fig 1].

2. Tidal marshes

Tidal marshes in South Korea are either mudflat or salt marsh ecosystems dominated by *Phragmites spp*. and *Suaeda spp*. Tidal marshes spanned approximately 248,200ha in 2018, down from 320,350 ha in 1999 [11]. 83% of these tidal marshes are located on the west coast and 17% on the south [12]. This is larger than the global estimate from Worthington, Spalding [13] that suggested 181,000 ha of tidal marsh extent for South Korea. Between 1987 and 2008, saltmarshes experienced a 20% decline in tidal marsh due to reclamation and landfill [Table 1].

There have been some salt marsh restoration efforts that have been undertaken with limited success [14]. Current tidal marsh carbon stocks are estimated to be 13,142,149 t C with carbon sequestration of 261,976 t CO_2e^{-1} yr^{-1} [6]. Based on these estimates from Lee, Kim [6], restoring 10% of South Koreas tidal marsh current habitat extent, would increase carbon sequestration by 26,198 t CO_2e^{-1} yr^{-1}. Restoration of saltmarshes in South Korea has already been successfully achieved by Koo, Je [14] The results from Lee, Kim [6] were similar to the mean carbon storage in South Korea's saltmarshes of between 146 and 255 t C ha^{-2}, with mud flats ranging from 182 to 286 t C ha^{-2} [15].

Mudflats are a separate ecosystem which contributes to coastal carbon sequestration [16,17] and have experienced steep global losses of 16% over the past four decades [18], and yet are not typically included as part of the blue carbon ecosystems. As a result, management actions to avoid mudflat losses and to enhance mudflat carbon sequestration have not yet been explored. Mudflats receive high fluxes of allochthonous organic carbon because of their high particle trapping capacity, with high rates of carbon preservation and low remineralisation rates [19,20]. Increased sediment runoff from

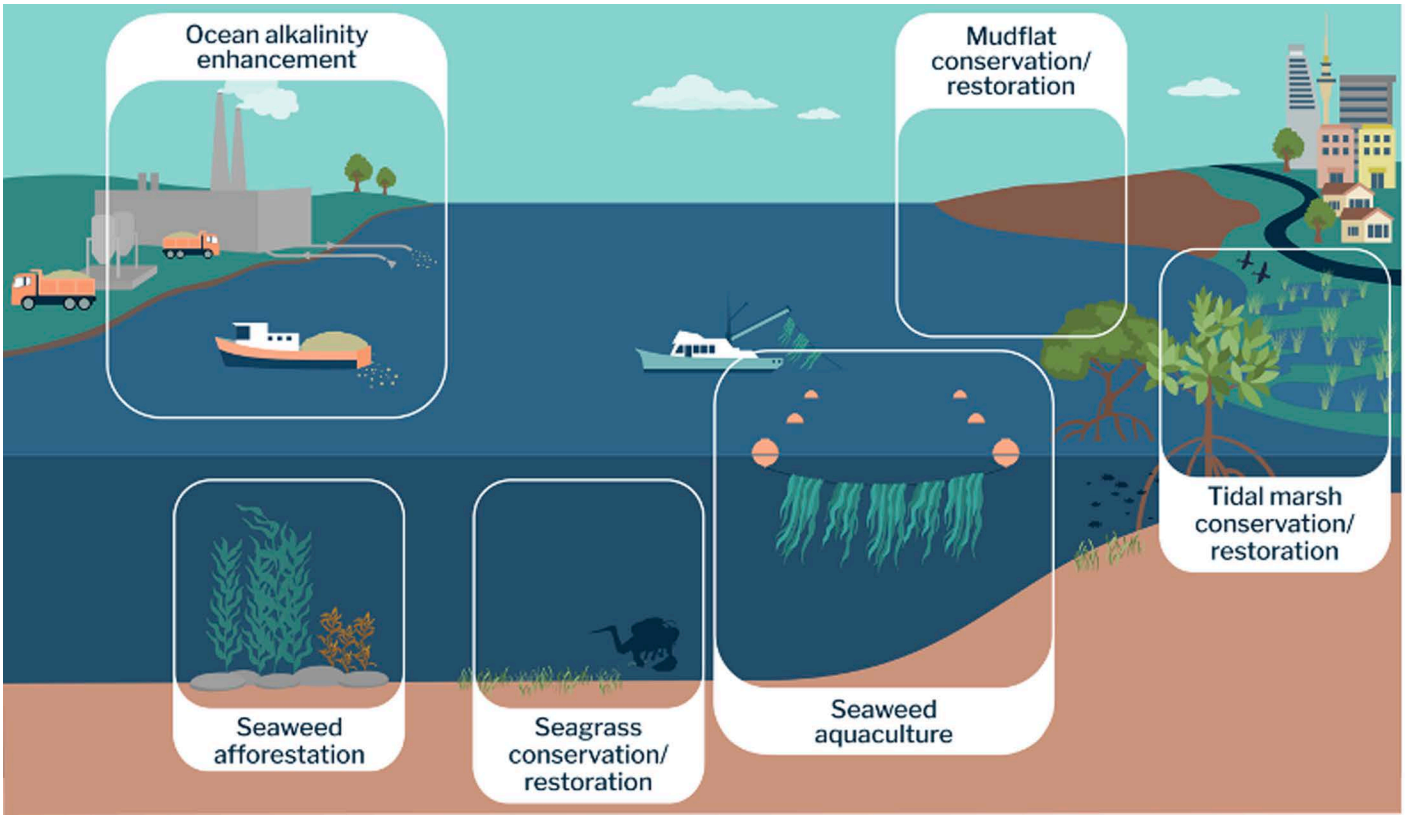

Fig 1. Overview of South Korea marine carbon sequestration opportunities, illustrated by Stacey McCormack, Visual Knowledge Pty Ltd.

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Table 1. Area of tidal marshes in Korea [11]. Between 1998 and 2003, a new more accurate remote sensing method was used, which is likely to be the reason for an increase in extent.

Year	1987	1998	2003	2008	2013	2018
ha	320,350	239,800	255,020	248,940	248,720	248,200

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industrial and agricultural activities on coastal watersheds may have increased the carbon storage and sequestration by mudflats. Preserving mudflats from degradation to avoid emissions is an important strategy that should be considered alongside the potential to increase mudflat carbon sequestration in South Korea, and explored more broadly globally as a new emerging component of blue carbon strategies.

3. Seagrasses

Seagrasses form intertidal and subtidal, down to 15 m depth in South Korea, vegetated coastal habitats with significant global potential for carbon sequestration and strong biodiversity co-benefits [21]. In South Korea there are nine seagrass species of four genera, including five *Zostera* spp, *with Zostera marina* being the dominant species. Six of these species are protected as “Marine Organisms under Protection” and 7 of the 15 IUCN Red List seagrass species occur in coastal waters of South Korea. Seagrass habitats are currently facing degradation in South Korea from mechanical impacts from harvesting of bivalves [22].

There is an estimated cover of 724,103 ha of seagrass in South Korea [Macreadie et al. 2021, Supp material page 29]. However, 50% of the historical seagrass cover has been lost due to land reclamation, eutrophication, aquaculture, dredging, and fisheries activities [3]. In a global estimate of seagrass carbon stocks Macreadie, Costa [7] estimated with relatively high uncertainty current carbon stocks of 165,800,000 t C in seagrass meadows, Supp material Table 4 Macreadie, Costa [7]. This differs from estimates from by Lee and Lee [23] who estimated the habitat area of *Zostera marina* in Korea to be between 55 and 70 km^2, which they assess to be about half of the historical area [3]. There are currently no estimates of carbon sequestration rates by South Korea's seagrass ecosystems, so we assume seagrasses sequester around 1.38 t CO_2 ha yr based on global averages [24]. Therefore, if degraded seagrass habitat was protected, avoiding emissions from 1% of South Korea's seagrass per year [0.01 x 165,800,000] and assuming that 50% of carbon the top 1m is lost in degradation [0.5 = 829,000], then 3,042,430 t C yr could be protected from being lost. The Kumning-Montreal global biodiversity framework which South Korea are aligned to requires no habitat loss, including from seagrasses by 2030. If we assumed that 10% of seagrass meadows were restored with a global average sequestration rate of 1.38 t CO_2 ha yr, this could sequester 999,262 t CO_2 yr from restoring seagrass habitats.habits

Table 2 provides a summary of blue carbon assets and opportunities in South Korea for saltmarshes and seagrasses, which are the two traditionally-recognised blue carbon habitats.

4. Mangroves

Mangroves do not occur naturally in South Korea. However, there are suggestions that Indonesian mangroves could be grown in Jeju – South Korea's largest island – where semi-mangrove species *Palliurus ramosissimus* are already established and starting to expand their area. We are unaware of any historical accounts of mangroves on the South Korean coastline and suggest extensive research around the risks of mangrove invasion should be undertaken before mangrove forestation is considered. Hence, we suggest that mangroves should not be introduced until evidence of lack of risks is available, given they are a non-native species, particularly since Jeju is a UNESCO Biosphere Reserve.

5. Seaweed

Seaweeds [macroalgae] is currently not recognised as one of the three blue carbon habitats because it generally does not sequester carbon at the growth site, but are considered as an emerging blue carbon habitat due to their potential for carbon sequestration [25]. Seaweed carbon sequestration requires, where seaweed grows on rocky substrate, seaweed fragments or dissolved organic products to break off and be exported to local or offshore sink depositional sites [9], which carries many challenges for carbon accounting such as carbon traceability and attribution [25]. However there has been significant discussion on the need to recognise seaweed blue carbon further [9,25], as evidence of significant contributions to carbon sequestration increase [26]. Indeed, seaweeds are already part of the blue carbon strategy of Japan and China, attesting to the traditional culture of using seaweed in Asia, which also applies to South Korea. Yet, estimates of the current area and loss rates of South Korean algal forests are currently lacking, which represents a gap that needs to

Table 2. Summary of carbon sequestration opportunities in South Korea for recognised blue carbon habitats.

Habitat	Current extent [ha]	Current Carbon stock [t/C]	Carbon sequestration [t/CO_2/ha/yr]	Current carbon sink size [t/CO_2 yr]	Available area for additional habitat [ha]	Potential additional carbon sequestration [t/CO_2 yr]
Saltmarsh	248,720	13,142,149	0.95	261,976	31,280*	482,316
Seagrasses	724,103	165,800,000	1.38	999,262	1,688,897	2,330,678

*Based on the difference between 1999 and 2014 tidal marsh extent.

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be bridged to assess the potential of seaweeds to contribute to South Korea's blue carbon and 'marine forest creation' targets [Table 3, Fig 2].

South Korea has extensive seaweed aquaculture [5], and has three of the 19 sites where seaweed carbon sequestration in sediment below the farms were assessed as part of a global study run by oceans 2050 [27]. Sustainable coastal seaweed aquaculture is an important contributor to carbon sequestration that has several co-benefits such as reducing ocean acidification and supporting rural communities [28]. South Korea has the largest wild seaweed restoration project in the world, the Sea Forestation project that has two decades of successful seaweed restoration, Fig 3 [5]. The Sea Forestation project has restored over 15,000 hectares of seaweed forest at an average cost of restoration: ~12,000 USD ha [https://kelpforestalliance.com/TNC-KFA-Kelp-Guidebook-2022.pdf]. For these reasons, South Korea is uniquely positioned to lead globally on seaweed carbon sequestration research and application.

The global coastline is 1.16 million km, kelp can be found in 25% of this area [30] or 290,000 km with a global seaweed NPP of 1.32 Pg C yr^{-1} [28]. Given the South Korean coastline is completely within the kelp biome and is 2,413 km long, a best estimate of South Korea's coastline seaweed NPP is 11 Mt C yr^{-1} [2,413/290,000 x 1,320,000,000]. Provided that, on average, 11% of this may be exported and reach a long-term sink [9], the carbon sequestration by South Korean seaweed may amount to 1.21 Tg C, implying non-additional carbon sequestration of 4.44 M/ CO_2 yr^{-1}. Assuming 10% of South Korea seaweed habitat may have been degraded, restoring this habitat would require restoring efforts along 241 km of coastline, leading to sequestration of an estimated 400,000 t CO_2 yr^{-1}.

South Korea also has a long history of seaweed farming. These farms deposit carbon in sediment beneath them which can be sequestered and measured [25,31]. Duarte, Delgado-Huertas [32] reported carbon burial rates at three South Korean farm sites. However, the farms included in the study were placed in advective environments that exported carbon rather than retain it on the seabed below the farm. Assuming a mean carbon burial with seaweed farming of about 1 t CO_2 ha yr^{-1} [32], it would take 1,362,000 hectares of ocean area in depositional environments to meet South Korea's blue carbon target, which is 2.4% of the South Korean EEZ. Seaweed farms also contribute to climate change mitigation by producing low carbon products [25,33] or displacing the use of fossil fuels in biofuel and other products [34]. For example, the methane-reducing seaweed *Asparagopsis* [35], low-carbon food resources relative to land options [33], and other products such as biofuels and bioplastics. However, the carbon footprint of most seaweed products and the emission abatement potential of seaweed carbon sequestration projects remains largely unquantified. There have also been suggestions to grow and sink seaweed directly into the deep sea, however, this carries serious risks and scientific uncertainties [36,37].

The Korean government's investment in kelp restoration is unique and has led to a large-scale, systematic approach to restoration. This approach has been underpinned by financial and logistical support over two decades and has shown impressive results. The project has demonstrated that large-scale restoration projects in the ocean

Table 3. Korean Fisheries and Research Agency 'marine forest creation strategy' [29].

	2009–2014	2015	2020	2030
Creation area [accumulation]	5,709 ha	7,000 ha	15,000 ha	35,000 ha
Strategy	- Marine forest creation. Seek and explore useful marine plants - Secure marine forest development site in advance - Promote the Marine Gardening day	- Seek and explore marine plants that can adapt to high water temperature - Explore and preserve natural ocean forests - Collaborate with environmental organisations. Expand the scale of the event	- Develop widespread, clear-water marine plants - Expand creation of low-cost highly-efficient marine forests - Globalize the marine gardening day	- Create ascidians coral reef marine forests - Provide space such as experience areas in order to enhance the happiness of people - Establish reforestation campaigns

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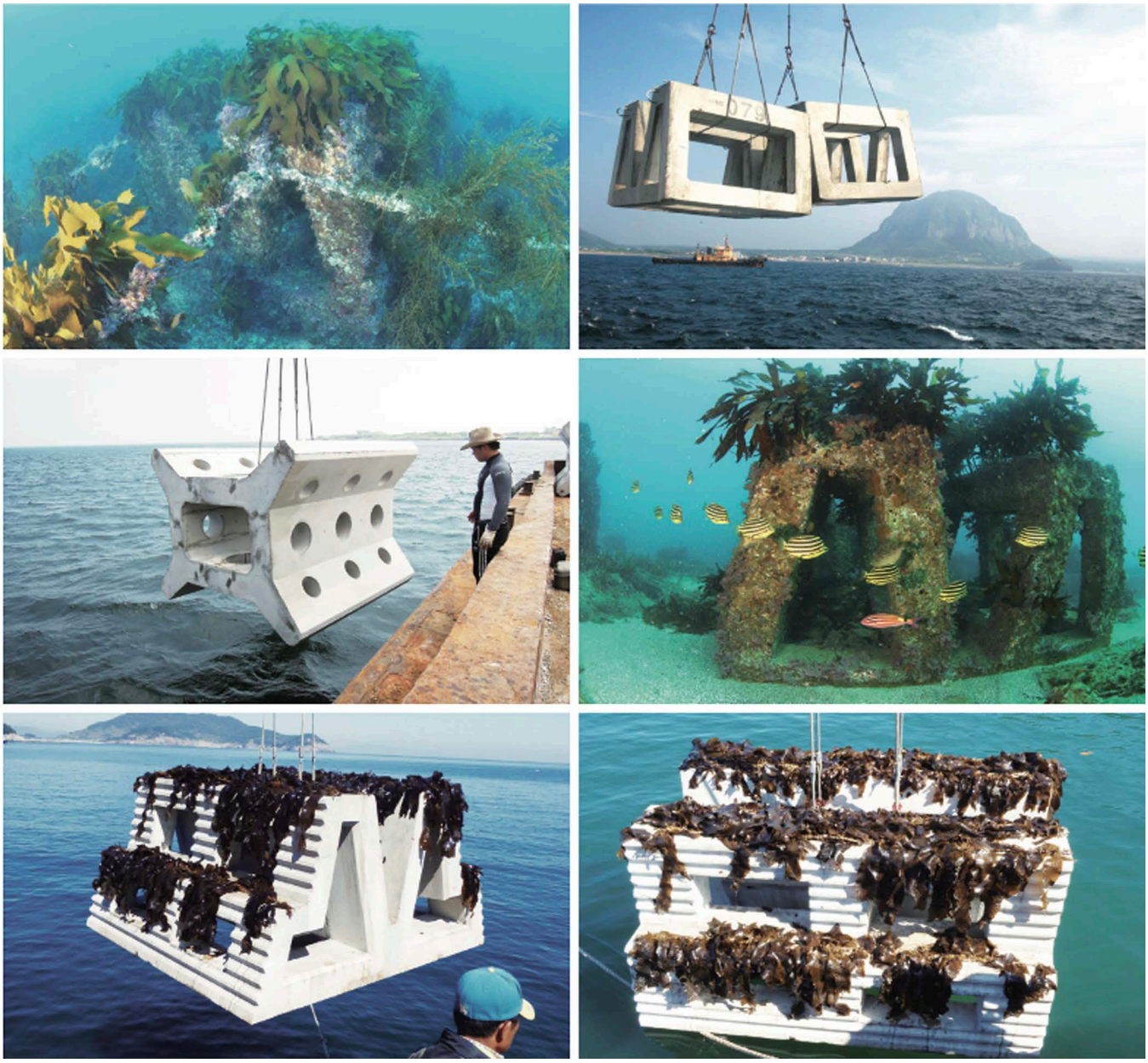

Fig 2. Marine forest creation map, reflecting the sites where artificial reef structures have been used for seaweed restoration attempts.

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are possible in South Korea given political support. We suggest a forensic carbon accounting method [38] be applied to seaweed carbon sequestration. This method can utilise novel forms of measuring seaweed carbon export such as using eDNA in deep sea sediment cores and modelling of fatty acids signatures to predict macroalgal carbon in marine sediments [39,40]. This method can use international collaboration to build from the J-Blue Japanese methodology for seaweed carbon credits [41], which has proved successful. This method should use the forensic carbon accounting framework developed by Hurd, Law [38], to assist seaweed projects to be recognised in South Korea's national carbon inventory.

Fig 3. Artificial reef construction for seaweed restoration, credit: Korean Fisheries and Research Agency [29].

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6. Marine Geoengineering

Marine geoengineering, is defined as "a deliberate intervention in the marine environment to manipulate natural processes, including to counteract anthropogenic climate change and/or its impacts"' [42]. There are many different proposals for marine geoengineering such as iron fertilisation to stimulate phytoplankton growth [43] and cloud seeding to reflect sunlight [44]. Each of the different proposals has its own unique risks, challenges and opportunities. They are also all at different stages of ideation with some having had no scientific testing and some having several major experiments done. With a large EEZ, South Korea should be prepared to be involved in international dialogues around marine geoengineering. In particular Ocean Alkalinity enhancement [OAE] which is one of the most promising marine geoengineering techniques [45], has relevance for South Korea given it can utilise a steel industry by product, steel slag, and South Korea has a large steel industry. Currently a carbon capture startup Capture6 [https://capture6.org/] has a proposed facility for carbon removal which includes OAE in South Korea. OAE is a carbon removal strategy that involves adding alkaline substances, such as steel slag, to seawater to shift the equilibrium of the carbonate system by displacing dissolved inorganic CO_2 in seawater into stable bicarbonates and carbonates. This reduces the concentration of CO_2 in seawater resulting in CO_2 uptake from the atmosphere. OAE also increases ocean pH which reduces ocean acidification. OAE has significant scalability potential to sequester millions of tonnes of CO_2 [46]. However, there are several potential impacts of weathering alkaline materials on marine life, although the net balance of positive and negative impacts remains unclear. For example, calcium or silica additions from alkaline materials could stimulate phytoplankton growth.

7. Other coastal carbon sequestration options

Seagrass and seaweed wrack

Accumulation of seagrass and seaweed wrack on beaches has become a global disposal challenge [47]. The annual CO_2-C flux from seagrass wrack globally is between 1.31 and 19.04 Tg C yr^{-1} [47]. However, if seagrass wrack cast is moved higher up along the shore it has been shown to have 72% lower emissions than wrack that was subjected to repeated wetting in the intertidal zone [47]. Seagrass wrack can be converted to biochar via pyrolysis with high efficiency, which can minimise emissions while sequestering carbon [48]. While we are not aware of examples of seagrass wrack locations in South Korea, given the extensive seagrass habitat it is likely seagrass wrack is an emission source that could be mitigated on the South Korean Coastline. Currently, *Sargassum horneri* wrack is extensive across South Korean coastlines and is collected in significant volumes [Fig 4, Table 4] with most of the collected biomass dumped in landfill.

Minimising dredging and bottom trawling

Dredging is a vital socio-economic activity that is needed to maintain safe navigable routes for shipping, vessel berths, marinas, and sheltered harbours around the world. In South Korea, approximately 729 million cubic meters of coastal dredging occurred between 2001 and 2008 [49]. Dredging often results in large quantities of sediment that must be disposed of – usually by either re-dispositioning dredged materials in subtidal environments or through placement in upland containment facilities, which could lead to greenhouse gas emissions [50,51]. Similarly, bottom trawling, a common commercial fishing method, displaces marine sediment, which renders organic carbon buried in sediments vulnerable to decomposition and CO_2 emissions [52, Atwood et al. 2024]. The mitigation of emissions from dredging and bottom trawling, are both activities that disturb the seafloor, are important components on avoidance emissions that should be quantified and managed in South Korea. Recently, an approach to use dredged sediments to support blue carbon restoration has been proposed as a method to reduce emissions and increase carbon sequestration [53]

8. Operationalising marine carbon sequestration in South Korea

Financing blue carbon is key to delivering habitat restoration [54], engaging private sector investment and creating jobs. Restoration efforts should also be environmentally sound and not in detriment of other parts of the South Korean coasts

Fig 4. ***Sargassum horneri*** **wrack collection Pohang beach, photographed by Peter Macreadie.**

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and oceans or local communities. Restoration efforts often encounter policy and cultural hurdles [55]. For example, unlike mudflats in Europe and other parts of the world, Korean mudflats have been utilized extensively by coastal residents with approximately 1,000 cultural entities operating in South Korean mudflat areas who may have different perspectives on blue carbon management [56].

Restoration is most cost-effective when projects are delivered at scale, with Kwon, Kim [57] suggesting that for all three types of blue carbon, the economic benefit of restoring 1 km^2 of each blue carbon type was at least five times greater than the cost of restoring it. However, much of this value corresponds to benefits that are not yet monetised, which weakens the business case to invest in blue carbon restoration projects. Carbon credits will be one of the key potential mechanisms to finance marine carbon sequestration in South Korea, with a number of potentially applicable methodologies for carbon credits in South Korea [Table 5], with methodologies for projects addressing mudflats still lacking.

9. Conclusion

This study demonstrates, South Korea's goal of sequestering 1,362,000 tonnes by ocean-based actions by 2050 is achievable. Restoring saltmarshes and seagrasses will lead to measurable actionable carbon sequestration, while seaweed restoration, seaweed aquaculture and mudflat protection will also provide valuable contributions. Similarly, emerging opportunities around marine geoengineering, managing seagrass and seaweed wrack and avoiding sediment disturbance

Table 4. Statistics on the annual and regional collection of *Sargassum horneri* along the coastline and at sea from Korea Marine Environment Information Portal [https://www.meis.go.kr/portal/main.do], MOF, Korea.

Year	Location	Jeollanam-do	Jeju-do [Jeju Island]	Chungcheongnam-do	Jeollabuk-do	Total [tonne]
2015	coastline	1,744	11,719	0	0	13,463
	sea	2,125	52	0	0	2,177
	subtotal	3,869	11,771	0	0	15,640
2016	coastline	0	295	0	0	295
	sea	0	19	0	0	19
	subtotal	0	314	0	0	314
2017	coastline	104	4,311	0	0	4,415
	sea	1,742	164	0	0	1,906
	subtotal	1,846	4,475	0	0	6,321
2018	coastline	1,455	1,951	0	0	3,406
	sea	430	19	0	0	449
	subtotal	1,885	1,970	0	0	3,855
2019	coastline	0	860	0	0	860
	sea	0	112	0	0	112
	subtotal	0	972	0	0	972
2020	coastline	1	3,654	1,150	0	4,805
	sea	0	1,533	0	0	1,533
	subtotal	1	5,187	1,150	0	6,338
2021	coastline	3,292	9,361	0	0	12,653
	sea	2,939	395	0	0	3,334
	subtotal	6,231	9,756	0	0	15,987
2022	coastline	91	412	755	4	1,262
	sea	79	0	0	0	79
	subtotal	170	412	755	4	1,341
Total [tonne]		14,002	34,857	1,905	4	50,768

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Table 5. Summary of applicable carbon credit methodologies for South Korea.

Habitat	Potentially applicable carbon credit methodologies
Saltmarsh	VM0033 Methodology for Tidal Wetland and Seagrass Restoration, v2.0
Mudflats	There are no current global methods for mudflat carbon, but the South Korean government has said it will mobilize all available means, such as strengthening international cooperation and expanding research and development, toward it having them recognised.
Seagrass	Carbon credits could be issued for avoided destruction and restoration of seagrass under VM00033 V2.0 released 4 September 2023.
Mangroves	VCS REDD Methodology Framework [VM0007]
Seaweed forests	J-Blue Japanese methodology, and Canopy Blue Western Australia method [https://canopyblue.co/purchase-kelp-reforestation-credits/].
Seaweed aquaculture	Oceans 2050 forthcoming method.
Ocean Alkalinity Enhancement	None yet available.

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by dredging can, if current scientific uncertainties are addressed, have important climate change mitigation contributions. This study is a first-pass estimate based on available data. We suggest that a national marine carbon strategy for South Korea should be developed and implemented to address current scientific and data gaps and generate investment into restoration trials and support restoration of coastal habitats at scale. Engaging industries within South Korea with high carbon emissions will help supplement government finance, as well as boost collaboration and innovation toward shared goals. While this study focuses on carbon, we note that many of the actions proposed - particularly those involving ecosystem restoration - will have other valuable 'ecosystem service' benefits, such as: biodiversity enhancement, coastal protection, and pollution removal, among others.

Author contributions

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Writing – review & editing: Finn Ross, Carlos M. Duarte, Ji Won Hong.

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